Settings tested: step from rest to 350 RPM, mean of repeats

ID	What changed	Kp	Ki	Filter α	PWM jitter (% duty)	vs baseline	Overshoot (%)	Rise 10–90 % (s)	Settling to ±5 % (s)	RPM std (RPM)	RPM std vs baseline	Power (W)	runs
A	Baseline (original design)	2.629	1.454	1.0 (off)	**0.177**	—	**4.2**	**5.46**	16.44	**3.16**	—	6.25	5
G44	Kp 2.23, α 0.2	2.235	1.454	0.2	**0.032**	−82 %	**4.3**	**5.39**	6.28	**4.36**	+38 %	6.23	1
G27	Kp 3.15, α 0.15	3.155	1.454	0.15	**0.034**	−81 %	**4.2**	**5.23**	7.07	**3.10**	−2 %	6.23	1
G35	α 0.2	2.629	1.454	0.2	**0.037**	−79 %	**4.3**	**5.36**	6.47	**3.22**	+2 %	6.33	1
G18	Kp 3.81, α 0.15	3.813	1.454	0.15	**0.041**	−77 %	**4.0**	**5.32**	7.95	**3.38**	+7 %	6.21	1
G17	Kp 3.81, α 0.2	3.813	1.454	0.2	**0.055**	−69 %	**3.7**	**5.51**	7.68	**3.12**	−1 %	6.30	1
G34	α 0.3	2.629	1.454	0.3	**0.056**	−68 %	**4.3**	**5.42**	6.46	**3.85**	+22 %	6.27	1
G25	Kp 3.15, α 0.3	3.155	1.454	0.3	**0.065**	−63 %	**4.7**	**5.43**	6.87	**3.67**	+16 %	6.10	1
G33	α 0.4	2.629	1.454	0.4	**0.066**	−63 %	**4.6**	**5.34**	6.90	**2.98**	−6 %	6.29	1
G40	Kp 2.23, α 0.6	2.235	1.454	0.6	**0.078**	−56 %	**4.5**	**5.44**	6.64	**3.30**	+4 %	6.26	1
G32	α 0.5	2.629	1.454	0.5	**0.083**	−53 %	**4.7**	**5.49**	6.57	**3.65**	+16 %	6.26	1
G24	Kp 3.15, α 0.4	3.155	1.454	0.4	**0.085**	−52 %	**3.9**	**5.46**	7.11	**3.41**	+8 %	6.23	1
G07	Kp 4.6, α 0.3	4.601	1.454	0.3	**0.090**	−49 %	**3.1**	**5.70**	8.40	**2.87**	−9 %	6.21	1
G31	α 0.6	2.629	1.454	0.6	**0.091**	−49 %	**3.9**	**5.42**	6.62	**3.36**	+6 %	6.26	1
G39	Kp 2.23, α 0.7	2.235	1.454	0.7	**0.094**	−47 %	**5.8**	**5.29**	8.90	**2.89**	−9 %	6.25	1
G23	Kp 3.15, α 0.5	3.155	1.454	0.5	**0.098**	−45 %	**3.8**	**5.43**	7.26	**3.30**	+4 %	6.22	1
G15	Kp 3.81, α 0.4	3.813	1.454	0.4	**0.099**	−44 %	**3.9**	**5.60**	7.58	**3.66**	+16 %	6.20	1
G14	Kp 3.81, α 0.5	3.813	1.454	0.5	**0.118**	−34 %	**3.8**	**5.46**	7.65	**3.06**	−3 %	6.13	1
G22	Kp 3.15, α 0.6	3.155	1.454	0.6	**0.121**	−32 %	**4.7**	**5.43**	7.19	**3.58**	+13 %	6.27	1
G29	α 0.8	2.629	1.454	0.8	**0.128**	−28 %	**4.3**	**5.33**	6.81	**2.92**	−8 %	6.19	1
G05	Kp 4.6, α 0.5	4.601	1.454	0.5	**0.135**	−24 %	**2.8**	**5.54**	8.57	**2.62**	−17 %	6.13	1

PWM jitter = RMS sample-to-sample change of the PWM command over the last 40 % of the run. Blue row = baseline. Row colour = startup overshoot: green ≤ 5 %, amber 5–15 %, red > 15 %.